

D 1 A

$$y = \begin{cases} \frac{\ln(1-x)}{2x} & \text{se } x < 0 \\ x^2 + 1/2 & \text{se } x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} \frac{\ln(1-x)}{2x} = \frac{\ln(1-x)}{x} \cdot \frac{1}{2} = \frac{-\ln(x-1)}{x} \cdot \frac{1}{2} = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} x^2 + \frac{1}{2} = \frac{1}{2}$$

$\frac{1}{2} \neq -\frac{1}{2}$ / discontinuità di 1ª specie

D 1 B

$$y = \begin{cases} \frac{\ln(1+x)}{2x} & \text{se } x < 0 \\ x^2 + \frac{1}{2} & \text{se } x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} \frac{\ln(x+1)}{2x} = \frac{\ln(x+1)}{x} \cdot \frac{1}{2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} x^2 + \frac{1}{2} = \frac{1}{2}$$

$\frac{1}{2} = \frac{1}{2}$ $x=0$ è punto di continuità

D 4 A

$$\begin{aligned} y &= x + e^x + \cos x & y(0) &= 0 + e^0 + \cos 0 = 2 \\ y' &= 1 + e^x - \sin x & y'(0) &= 1 + 1 - 0 = 2 \\ y'' &= e^x - \cos x & y''(0) &= 1 - 1 = 0 \end{aligned}$$

$$P(x) = 2 + 2x$$

D 4 B

$$y = -\frac{x^2}{2} + e^x + \sin(x) \quad y(0) = 1$$

$$y' = \frac{(-2x)(2)}{4} + e^x + \cos x = -\frac{4x}{4} + e^x + \cos x = -x + e^x + \cos x = 1 + 1 = 2$$

$$y'' = -1 + e^x - \sin(x) \quad y''(0) = -1 + e^0 = 0 \quad P(x) = 1 + 2x$$

D 5 A

$$\int_1^2 \frac{2e^x}{e^x+2} = \left[2 \ln(e^x+2) \right]_1^2 = \left(2 \ln(e^2+2) \right) - \left(2 \ln(e+2) \right) = \ln(e^2+2)^2 - \ln(e+2)^2 = \ln \left(\frac{(e^2+2)}{(e+2)} \right)^2$$

$$\int \frac{2e^x}{e^x+2} = \int \frac{2e^x}{e^x+2} \cdot \frac{1}{e^x} = \int \frac{2}{e^x+2} = \int \frac{2}{y} = 2 \int \frac{1}{y} = 2 \ln(y) = 2 \ln(e^x+2)$$

$$y = e^x$$

$$dy = e^x dx$$

$$\frac{dy}{e^x} = dx$$

D 5 B

$$\int_{1/2}^1 \frac{4e^{2x}}{e^{2x}+2} = \left[2 \ln(e^{2x}+2) \right]_{1/2}^1 = \left(2 \ln(e^2+2) \right) - \left(2 \ln(e+2) \right) = \ln(e^2+2)^2 - \ln(e+2)^2 = \ln \left(\frac{e^2+2}{e+2} \right)^2$$

$$\int \frac{4e^{2x}}{e^{2x}+2} = \frac{4e^{2x}}{e^{2x}+2} \cdot \frac{1}{2e^{2x}} = \int \frac{2}{y} = 2 \int \frac{1}{y} = 2 \ln(y) = 2 \ln(e^{2x}+2) + C$$

$$y = e^{2x} + 2$$

$$dy = 2e^{2x} dx$$

$$\frac{dy}{2e^{2x}} = dx$$

E 2 A

$$f(z) = 3$$

$$f'(z) = 4$$

$$f(x) = \sqrt{f^2(x) + 2}$$

$$f'(x) = \frac{1}{2\sqrt{f^2(x) + 2}} \cdot 2 \cdot f(x) \cdot f'(x)$$

$$f'(z) = \frac{1}{2\sqrt{9+2}} \cdot 2 \cdot 3 \cdot 4 = \frac{1}{4} \cdot 12 = 3$$

E 2 B

$$f(z) = 2$$

$$f'(z) = 3$$

$$f(x) = \sqrt{f^2(x) + 2}$$

$$f'(x) = \frac{1}{2\sqrt{f^2(x) + 2}} \cdot 2 \cdot f \cdot f'$$

$$f'(z) = \frac{1}{2\sqrt{4+2}} \cdot 2 \cdot 2 \cdot 3 = \frac{1}{3} \cdot 2 \cdot 3 = 2$$

D 7 A

$$\lim_{n \rightarrow +\infty} \frac{n \ln^3(n) - \sqrt{n} + n^{3/2}}{2n + 3\sqrt[3]{n} - n \ln^4(n)} = \frac{n \left(\ln^3(n) - \frac{\sqrt{n}}{n} + \frac{n^{3/2}}{n} \right)}{n \left(2 + \frac{3\sqrt[3]{n}}{n} - \ln^4(n) \right)} = \frac{\ln^3(n) + n^{1/2}}{-\ln^4(n) + 2} \sim \frac{\sqrt{n}}{-\ln^4(n)} = -\infty$$

D 7 B

$$\lim_{n \rightarrow +\infty} \frac{n \ln^3(n) - \sqrt{n} - n^{1/3}}{2n^2 + 3\sqrt[3]{n} - n \ln^4(n)} = 0 \quad 2n^2 \gg \text{numerator}$$

Eg 1

$$y = \frac{e^x}{x^2 - 1}$$

DOMINIO

$$x^2 - 1 \neq 0 \quad x^2 \neq 1 \quad x \neq \pm 1 \quad D = \mathbb{R} \setminus \{-1, 1\}$$

LIMITI

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - 1} = +\infty \quad \text{ess. obliquo?}$$

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - 1} \cdot \frac{1}{x} = \frac{e^x}{x^3 - x} = \frac{e^x}{x(x^2 - 1)} = +\infty \quad \text{semplice infinito}$$

$$\lim_{x \rightarrow -\infty} \frac{e^x}{x^2 - 1} = 0 \quad \text{ess. orizzontale}$$

$$\lim_{x \rightarrow -1^-} \frac{e^{-1}}{1 - 1} = +\infty$$

$$\lim_{x \rightarrow -1^+} \frac{e}{1 - 1} = -\infty$$

$$\lim_{x \rightarrow 1^-} \frac{e^x}{x^2 - 1} = \frac{e}{1 - 1} = \frac{e}{0^-} = -\infty$$

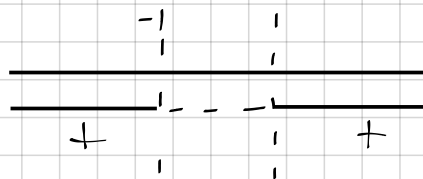
$$\lim_{x \rightarrow 1^+} \frac{e^x}{x^2 - 1} = \frac{e^x}{1^+ - 1} = \frac{e}{0^+} = +\infty$$

ess. verticale

SEGNO

$$\frac{e^x}{x^2 - 1} > 0 \Rightarrow \begin{cases} e^x > 0 \\ x^2 - 1 > 0 \end{cases} \begin{cases} \forall x \\ x < -1 \vee x > 1 \end{cases}$$

$$\neq 0 \quad \text{per } x \in (-\infty, -1) \cup (1, +\infty)$$

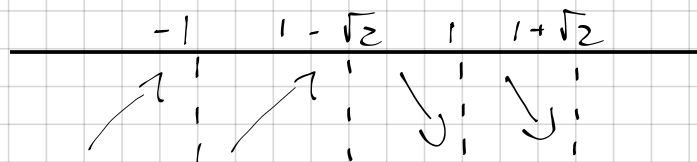


MONOTONIA

$$y = \frac{e^x}{x^2 - 1}$$

$$y' = \frac{(e^x)(x^2 - 1) - (e^x)(2x)}{(x^2 - 1)^2} = \frac{x^2 e^x - e^x - 2x e^x}{(x^2 - 1)^2}$$

$$\frac{x^2 e^x - e^x - 2x e^x}{(x^2 - 1)^2} > 0 \quad \left\{ \begin{array}{l} e^x (x^2 - 2x - 1) > 0 \\ (x^2 - 1)^2 > 0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} e^x > 0 \\ x^2 - 2x - 1 > 0 \\ x \neq \pm 1 \end{array} \right. \left\{ \begin{array}{l} \forall x \\ x < 1 - \sqrt{2} \vee x > 1 + \sqrt{2} \\ x \neq \pm 1 \end{array} \right.$$



MAX & MIN

$$\frac{x^2 e^x - e^x - 2x e^x}{(x^2 - 1)^2} = 0$$

$$x^2 e^x - e^x - 2x e^x = 0$$

$$e^x (x^2 - 2x - 1) = 0$$

$$x^2 - 2x - 1 = 0$$

$$x_1 = 1 - \sqrt{2} \quad x_2 = 1 + \sqrt{2}$$

x_1 punto di max relativo

x_2 punto di min relativo